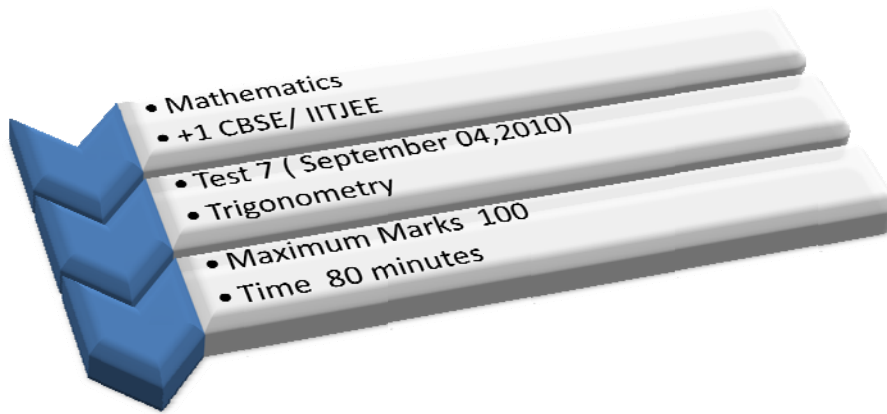


Name
Class
R. No.



Instructions

1. The test contains 10 questions.

Prove the following identities (Q1-Q9)

9 × 10 Marks

Q1:
$$\frac{(\sin 2\alpha - \sin 6\alpha) + (\cos 2\alpha - \cos 6\alpha)}{\sin 4\alpha - \cos 4\alpha} = 2 \sin 2\alpha$$

Q2: $\cos^2 \alpha - \sin^2 2\alpha = \cos^2 \alpha \cos 2\alpha - 2 \sin^2 \alpha \cos^2 \alpha$

Q3:
$$\frac{1 - 2 \cos^2 \alpha}{2 \tan\left(\alpha - \frac{\pi}{4}\right) \sin^2\left(\frac{\pi}{4} + \alpha\right)} = 1$$

Q4: $1 + 2 \cos 7a = \frac{\sin 10.5a}{\sin 3.5a}$

Q5: $\frac{\sqrt{1+\cos \alpha} + \sqrt{1-\cos \alpha}}{\sqrt{1+\cos \alpha} - \sqrt{1-\cos \alpha}} = \cot\left(\frac{\alpha}{2} + \frac{\pi}{4}\right), \pi < \alpha < 2\pi$

Q6: $2(\sin^6 \alpha + \cos^6 \alpha) - 3(\sin^4 \alpha + \cos^4 \alpha) + 1 = 0$

Q7: $\frac{1 + \tan \alpha + \tan^2 \alpha}{1 + \cot \alpha + \cot^2 \alpha} = \tan^2 \alpha$

Q8: $\frac{\sqrt{2} - \sin \alpha - \cos \alpha}{\sin \alpha - \cos \alpha} = \tan \left(\frac{\alpha}{2} - \frac{\pi}{8} \right)$

Q9: $\frac{2 \cos^2 2\alpha + \sqrt{3} \sin 4\alpha - 1}{2 \sin^2 2\alpha + \sqrt{3} \sin 4\alpha - 1} = \frac{\sin(4\alpha + 30^\circ)}{\sin(4\alpha - 30^\circ)}$

Q10: Simplify the following expressions

9 × 10 Marks

a) $\frac{1 + \sin 2x}{(\sin x + \cos x)^2}$

b) $\cos 0 + \cos \frac{\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{5\pi}{7} + \cos \frac{6\pi}{7}$